Computer Science Department

A Template for Senior Project Kickoff Document – Fall 2025

Title of the project:	Smart Campus Navigation with Real-Time Positioning and Adaptive Accessibility Routing
Team Members:	Aiym Abay, Madina Dairova, Zhansaya Zhassulanova, Aiymzhan Ashirbek, Aruzhan Ryssayeva
Project Advisor/Co-Advisors	Askar Boranbayev

Motivation and Summary (20%)

The project proposed addresses the challenges of navigating large university campuses. Existing navigation solutions, such as Google Maps, have been found inadequate due to the lack of indoor accuracy, insufficient real-time updates, and the absence of campus-specific integrations. These shortcomings have resulted in inefficiencies, particularly in locating classrooms, avoiding congested areas, and accessing essential university services.

To mitigate these challenges, a Smart Campus Navigation and Lifestyle Platform will be developed. The system will combine real-time positioning, accessibility-aware routing, and campus service integration, ensuring a seamless and adaptive navigation experience for students, faculty, and visitors.

Inputs to the system will include:

- GPS signals for outdoor positioning.
- User preferences (e.g., shorter route, stair-free accessibility).
- Campus event and construction data for real-time rerouting.
- Academic data such as class schedules for context-aware navigation.
- Accessibility requirements for users with mobility challenges.

Outputs of the system will include:

- A cross-platform mobile and web application offering real-time navigation, accessibility tools, and academic planning.
- AI-driven recommendations, including study space availability, dining options, and event notifications.
- Academic tools such as GPA calculators, planners, and calendar integrations.
- A secure platform for student discounts and loyalty services.

The proposed solution will rely on hybrid positioning technologies, AI-based optimization, and cloud-based data processing. The platform will be designed for scalability to support thousands of daily users.

System Description (20%)

The Smart Campus Navigation system will be built around four core components, developed across two semesters to ensure steady progress and integration.

1. **Mobile Application:**
   The mobile app will act as the main interface for students, faculty, and visitors. Designed for both iOS and Android, it will provide easy-to-use navigation with AR overlays, reminders, and review features. It will also include helpful academic tools such as GPA calculators and planners, making it more than just a navigation tool.
2. **Positioning & Navigation Engine:**
   At the heart of the system will be a hybrid positioning engine powered by GPS. It will generate accurate routes that adjust to individual needs—whether that means finding the fastest path, choosing accessible routes, or avoiding construction zones.
3. **AI & Analytics Backend:**
   A cloud-based backend will handle data collection and analysis. By drawing on multiple sources, it will provide personalized recommendations for study spots, dining options, and campus events. Over time, feedback from users will be used to refine and improve these suggestions.
4. **Integration Layer:**
   To complete the experience, the system will connect seamlessly with external services such as Google Calendar, university databases, and student discount platforms. This will allow schedules, events, and perks to be integrated directly into the app.

Together, these components will operate in real time. Data will be processed in the backend and immediately reflected in the app, creating a smooth and adaptive experience for every user.

Hardware and Software Requirements (10%)

Hardware:

- User testing and deployment of Smartphones (iOS and Android).
- Backend services and scalability through cloud hosting environment (AWS, Azure, or Google Cloud).

Software:

- Android/iOS compatibility with a cross-platform mobile development framework (Flutter or React Native).
- AR SDKs (ARCore, ARKit) for augmented reality direction.
- Real-time updates, authentication, and data storage in the form of a backend (Firebase or AWS).
- Databases: MongoDB is used to store unstructured review data, and PostgreSQL is used to store structured academic and discount information.
- Machine Learning Model: TensorFlow Lite and NLP Overview of Sentiment Analysis in Reviews.
- External discount provider and Google Calendar secure API integrations.
- Development and testing tools (GitHub, Docker, CI/CD pipeline, project management via JIRA/Trello).

Evaluation Criteria and Plan (10%)

The evaluation of the Smart Campus Navigation app will be based on both quantitative and qualitative scales of user satisfaction.

- **Navigation Accuracy:**
  - Checking the system works in different parts of the university to achieve the uncertainty of 2 meters
  - Comparing our findings to existing solutions (for instance, 2GIS)
- **System Performance:**
  - Latency: The typical response time for route calculation and review submission should be less than 2 seconds
  - Scalability: the capacity to manage at least 1,000 users at once during heavy workload (peak hours)
- **Review & Sentiment Analysis Reliability:**
  - Measures of recall and precision for sentiment classification using natural language processing
  - AI-generated insights and reviews with human labels are compared
- **Discount & Rewards System:**
  - Discount transaction validation using a secure API
  - Fraud detection rate and user reward distribution dependability
- **Accessibility & Inclusivity:**
  - Usability tests with students who need routes that are accessible by wheelchair
  - Support for many languages has been confirmed with overseas students
- **User Experience and Adoption:**
  - Measures of adoption rates, perceived utility, and student satisfaction through surveys and interviews
  - Analytics dashboard monitoring navigation, reviews, and sales engagement
  - **Testing Procedures:**
  - Unit Testing: Individual module verification
  - Integration Testing: Ensuring seamless front-end, back-end, and API communication
  - Functional Testing: Confirming that things function as planned in practical situations
  - Acceptance Testing: Review of end users by student volunteers
  - Performance Testing: Stress testing with a large number of concurrent users

Risks and Contingency Plans (10%)

As our system is expected to achieve indoor positioning accuracy of ≤2 meters, the breakdown of beacon networks would greatly inhibit this fundamental capability. Our fallback plan includes the use of an alternative indoor positioning method utilizing Wi-Fi fingerprinting in conjunction with sensor fusion via device accelerometers, gyroscopes, and magnetometers. Although the fallback method may lower expectations for accuracy to ≤5 meters, the method preserves necessary functionality without the need for the installation of additional hardware infrastructure.

Since our hybrid localization engine is Wi-Fi triangulation-based, network access constraints or network instability can incapacitate the positioning engine. To address this vulnerability, we will develop an offline-first design with cached maps and pre-computed routes in a way such that basic navigating persists even without network access.

The AR component also exhibits compatibility hazards across varied device ecosystems. The AR platforms like ARCore and ARKit may experience compatibility issues in certain devices or operating system versions and therefore render the AR-based navigation interface inoperable. Our fallback mechanism employs progressive enhancement in such a way that 2D overlay-based navigation acts as the fallback when the device does not support AR. Furthermore, we will develop web-based AR utilizing the WebXR technologies for broader device support and design simplified visual guidance systems using standard camera views in combination with directional cues.

Our system's artificial intelligence and machine learning parts also carry performance uncertainties that can affect user experience. In case the accuracy of sentiment analysis is not good or crowd prediction algorithms work badly, we use rule-based fallback systems. Machine learning strategies for crowd prediction will be replaced by statistical models wherever necessary, and human curation systems integrated with voting mechanisms by the communities will act as fallback for AI-based recommendations.

Third-party API dependencies cause vulnerability outside of our own control. Integration malfunctions for Google Calendar, discount verification centers, or university course systems can cause core functionality disruptions. Our planning for contingencies includes the development of local storage for important data, the development of manual entry substitutes for calendar events, and the use of partner-based discount systems integrated with local verification systems.

University access restrictions for data might hinder integration with official building maps, room schedules, or course offerings. Our approach consists of crowdsource mapping technologies in which the user provides building resources, collaboration with student groups in gathering data, and use of publicly distributed campus maps and OCR scanning technologies for digitization.

Privacy and security compliance concerns need proactive risk management. Location tracking for the user, handling review data, and processing personal information need compliance with dynamic regulations for privacy. Our contingency strategy employs anonymous use modes with limited functionality, deploys differential techniques for the aggregation of data, and generates complete opt-outs in the build while preserving essential navigation functions.

Navigation update delays, AR overlay lag, or slow route optimization could render the system impractical for daily campus use. Our mitigation strategy emphasizes client-side processing for critical features, edge computing implementation for reduced latency, pre-computed route caching systems, and progressive loading with placeholder content to maintain perceived performance.

Deliverables (10%)

Hybrid Positioning Engine - GPS-based outdoor positioning module achieving ±3 meter accuracy, Wi-Fi triangulation algorithm implementation with comprehensive signal strength mapping, BLE beacon integration system with proximity detection capabilities, and seamless indoor/outdoor transition logic with sophisticated zone detection algorithms

Cross-Platform Mobile Application Foundation - Complete Flutter/React Native framework setup with modular architecture, basic UI/UX implementation following Material Design and iOS Human Interface Guidelines, secure user authentication system with token management, and core navigation interface with advanced map integration and route display capabilities

Intelligent Route Planning System - Multi-criteria route optimization engine supporting shortest, accessible, and least crowded pathways, real-time crowd density prediction using historical data and machine learning models, dynamic rerouting algorithms responding to live campus events, and accessibility route calculation prioritizing wheelchair-friendly pathways and elevator access

Augmented Reality Navigation Interface - Complete ARCore/ARKit integration with camera-based positioning, real-time AR overlay system featuring directional arrows and landmark recognition, visual landmark detection and annotation system, and performance optimization maintaining 60fps target for sustained AR usage

Comprehensive Review and Feedback Platform - Multi-category review system covering cafés, classrooms, study spaces, and facilities, advanced NLP sentiment analysis engine with custom campus-specific training data, weighted reputation algorithm with anti-spam mechanisms, and review aggregation with trending analysis using temporal weighting

AI-Driven Recommendation Engine - Personalized study spot recommendations based on user preferences and historical data, dining suggestion system accommodating dietary restrictions with preference learning, event and activity recommendation with social integration, and context-aware suggestions based on location, time, and behavioral patterns

Secure Discount and Rewards Platform - External discount provider API integration with comprehensive fraud prevention, student verification system with university ID validation, loyalty point system tied to review engagement and app usage, and secure payment processing integration for premium features

Academic Tools Integration Suite - Comprehensive GPA calculator with grade prediction algorithms, academic planning tool with degree requirement tracking, bidirectional Google Calendar synchronization with smart scheduling, and study session planning with location-based recommendations

Robust Backend Services Architecture - Microservices-based API system with Docker containerization, real-time data processing pipeline using Apache Kafka/Redis, secure user authentication with OAuth 2.0 and JWT implementation, and database optimization with comprehensive caching layers

Comprehensive Analytics Dashboard - Real-time usage metrics visualization with interactive charts, user behavior analysis with conversion funnels and retention tracking, review sentiment trends analysis with automated reporting, and system performance monitoring with alerting mechanisms

Security and Privacy Framework - End-to-end encryption for sensitive user data, GDPR compliance toolkit with data export and deletion capabilities, anonymous usage tracking with differential privacy implementation, and security audit tools with penetration testing documentation

Production-Ready Mobile Applications - iOS App Store submission with comprehensive TestFlight beta testing program, Google Play Store deployment with staged rollout capabilities, Progressive Web App version for broader accessibility, and offline functionality with local data synchronization

Complete Documentation and Deployment Package - Technical documentation with detailed API specifications and architectural diagrams, user manual and training materials for campus administrators, installation and maintenance guides for IT departments, and open-source components with proper licensing and contribution guideline

Project Management and Tasking (10%)

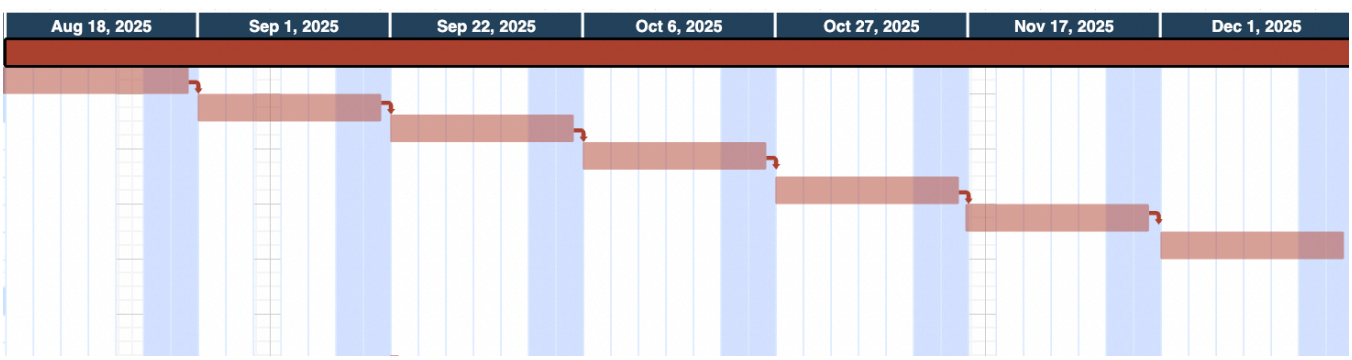
We are planning to use **Notion** as the project management tool, as it best suits our needs for planning tasks, creating shared documentation, and tracking progress. It also allows integration with Kanban boards and

Gantt charts, which will be useful for organizing work. In addition, we will use **GitHub Projects** for technical task tracking and version control.

The principal tasks planned for this project are divided into two semesters. In the **Fall 2025 semester**, our team will focus on research, requirements gathering, technology familiarization, system architecture, and initial prototype development. In the **Spring 2026 semester**, the project will shift towards full feature integration, testing, optimization, and deployment.

Fall 2025

	Task Name	Duration	Start	ETA
1	Project Kick-off & Planning	2 weeks	Aug 18, 2025	Aug 31, 2025
2	Requirement Gathering & Research	3 weeks	Sep 1, 2025	Sep 21, 2025
3	Technology Familiarization	2 weeks	Sep 22, 2025	Oct 5, 2025
4	System Architecture & Design	3 weeks	Oct 6, 2025	Oct 26, 2025
5	Prototype Development – Phase 1	3 weeks	Oct 27, 2025	Nov 16, 2025
6	Prototype Development – Phase 2	2 weeks	Nov 17, 2025	Nov 30, 2025
7	Documentation & Mid-term Evaluation	1 week	Dec 1, 2025	Dec 7, 2025



Spring 2025 (tentative)

- Full integration of remaining features (discount system, academic tools, accessibility options).
- Testing & Evaluation: indoor/outdoor navigation accuracy, AR usability, and sentiment analysis.
- Performance optimization for scalability, latency, and security.
- Development of Analytics Dashboard for adoption metrics and review trends.
- Final documentation, evaluation report, and presentation.
- Deployment of mobile app (iOS/Android) and backend services.

References and Further Reading (10%)

[1] MarketsandMarkets. (2023). *Indoor Location Market by Component, Application, and Region – Global Forecast to 2028.*
[2] ResearchAndMarkets. (2024). *Global Smart Campus Market – Growth, Trends, COVID-19 Impact, and Forecasts (2024–2029).*
[3] Grand View Research. (2023). *Indoor Positioning and Navigation Market Size, Share & Trends Analysis*

Report.
[4] EDUCAUSE Review. (2022). *The Rise of the Smart Campus: Implications for Higher Education IT.*